

Website Phishing Detection

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Objectives

- Detect Phishing websites
- Identify important variables
- Build classification models that predicts the least amount of false negatives





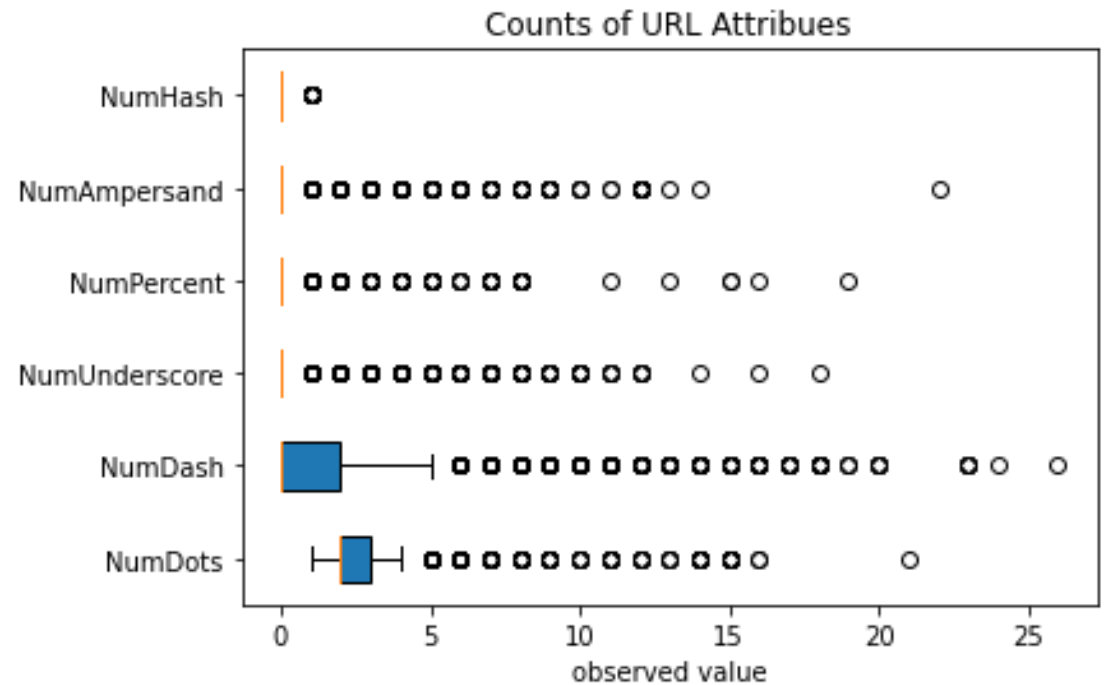
Data Description

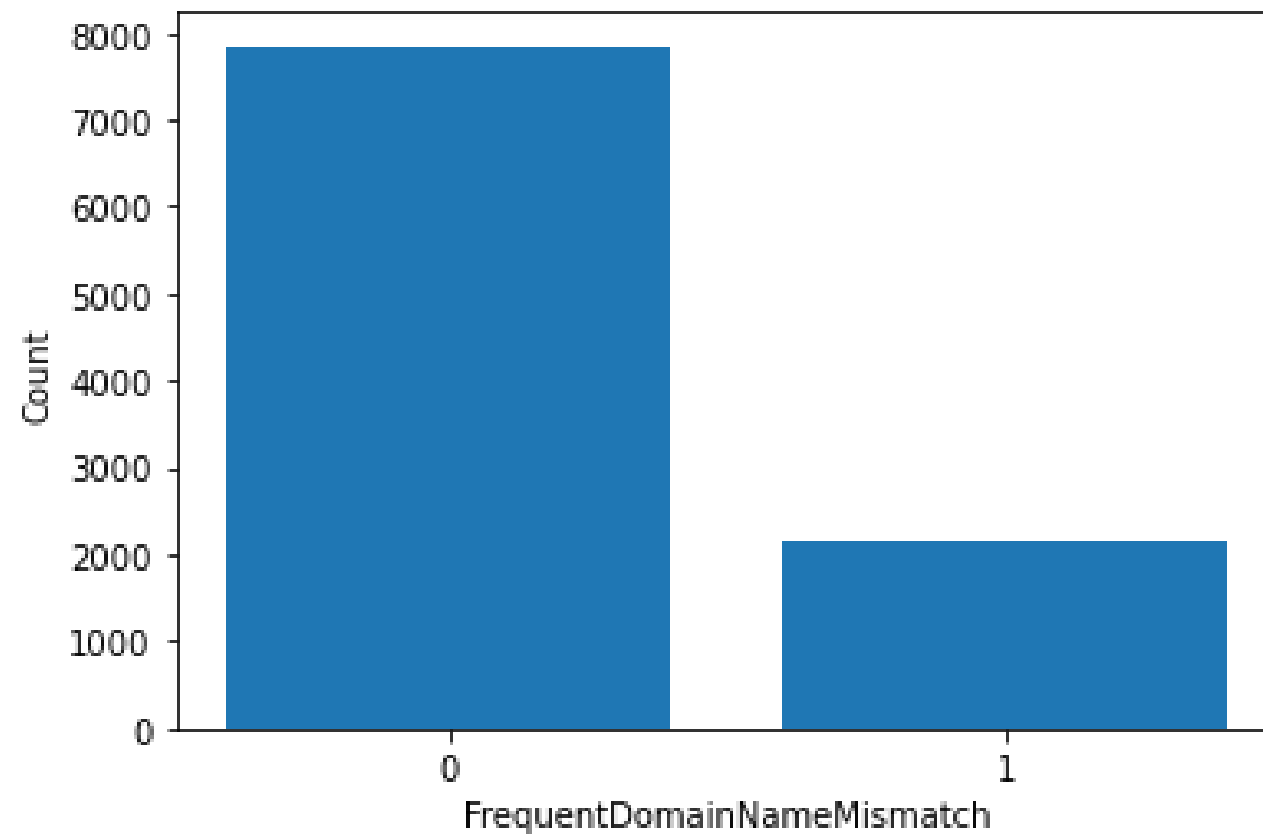
- Collected from January to May 2015 and from May to June 2017
- 48 features
- 10,000 observations
 - 5000 phishing and 5000 legitimate web pages

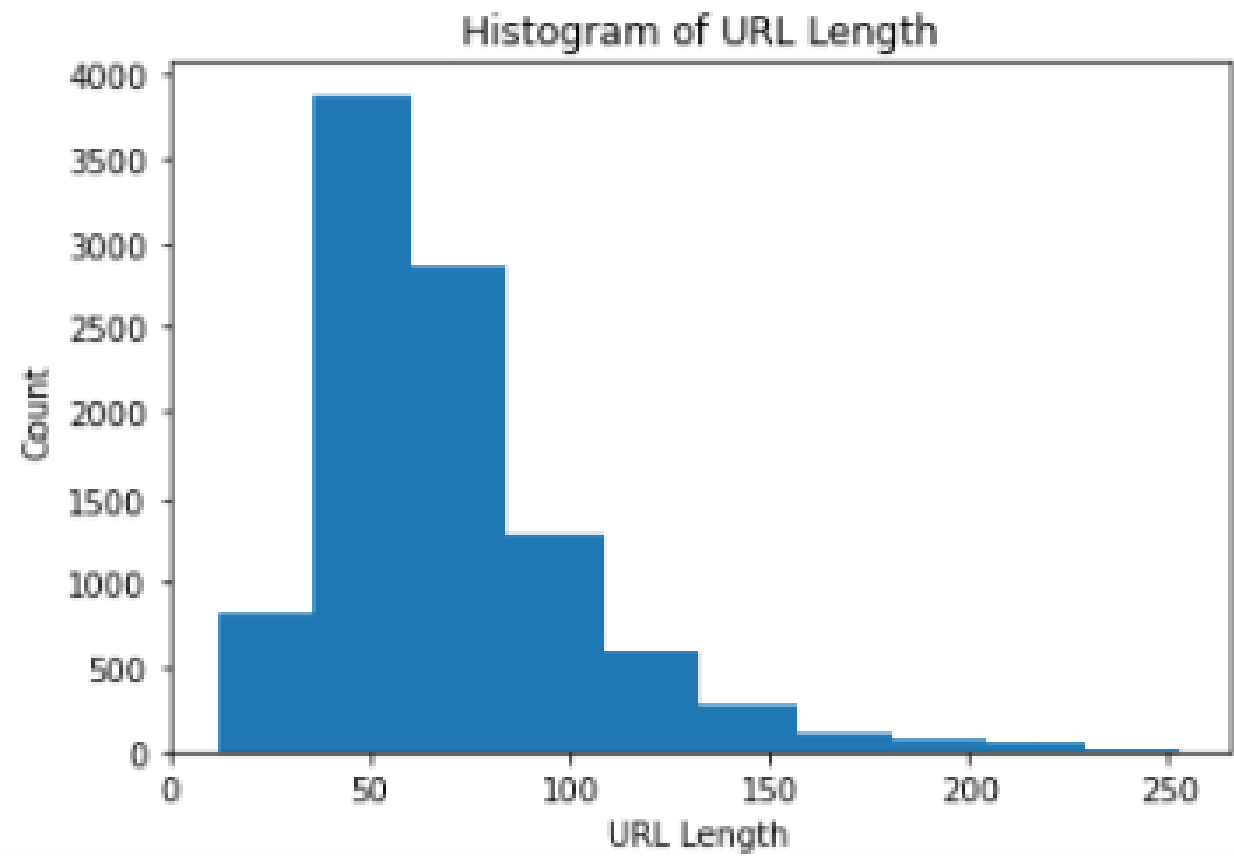
Column name	Data type	Description	NumPercent	Numerical	Number of “%” in URL
id	Categorical	Unique identifier	NumQueryComponents	Numerical	Number of query components
NumDots	Numerical	Number of “.” in URL	NumAmpersand	Numerical	Number of “&” in URL
SubdomainLevel	Numerical	Level of subdomains	NumHash	Numerical	Number of “#” in URL
PathLevel	Numerical	Level of path segments	NumNumericChars	Numerical	Number of Numeric characters
UrlLength	Numerical	Length of characters in URL	NoHttps	Categorical	Whether Https is present in URL
NumDash	Numerical	Number of “-” in URL	RandomString	Categorical	Whether random string is present
NumDashInHostname	Numerical	Number of “-” in host name	IpAddress	Categorical	Whether an Ip address present
AtSymbol	Categorical	Whether “@” is present in URL	DomainInSubdomains	Categorical	Is domain in subdomains
TildeSymbol	Categorical	Whether “~” is present in URL	DomainInPaths	Categorical	Is domain in paths
NumUnderscore	Numerical	Number of “_” in URL	HttpsInHostname	Categorical	Is https in host name upon request
			HostnameLength	Numerical	Length of host name

PathLength	Numerical	Lenth of path	FakeLinkInStatusBar	Categorical	Is there a Fake Link in Status Bar
QueryLength	Numerical	Length of query	RightClickDisabled	Categorical	Is right click disabled
DoubleSlashInPath	Categorical	Is “//” in the path	PopUpWindow	Categorical	Are there pop-up windows
NumSensitiveWords	Numerical	Number of sensitive words	SubmitInfoToEmail	Categorical	Can info be submitted to email
EmbeddedBrandName	Categorical	Is brand name embedded	IframeOrFrame	Categorical	Is HTML Iframe or frame
PctExtHyperlinks	Numerical	Number of .pct extension hyperlinks	MissingTitle	Categorical	Is page missing title
PctExtResourceUrls	Numerical	Number of .pct extension Resource URLs	ImagesOnlyInForm	Categorical	Are images only in form
ExtFavicon	Categorical	Is a Favicon file extension present	SubdomainLevelRT	Categorical	Standardized Level of subdomains
InsecureForms	Categorical	Does page have insecure forms	UrlLengthRT	Numerical	Standardized Length of characters in URL
RelativeFormAction	Categorical	Does page have relative form action	PctExtResourceUrlsRT	Numerical	Standardized PctExtResourceUrlsRT
ExtFormAction	Categorical	Does page have form action extension	AbnormalExtFormActionR	Categorical	Standardized AbnormalExtFormActionR
AbnormalFormAction	Categorical	Does page have abnormal form action	ExtMetaScriptLinkRT	Categorical	Standardized ExtMetaScriptLinkRT
PctNullSelfRedirectHyperlink	Numerical	Null .pct redirecting hyperlink	PctExtNullSelfRedirectHyperlinksRT	Numerical	Standardized PctExtNullSelfRedirectHyperlinksRT
FrequentDomainNameMismatch	Categorical	Is there Frequent Domain Mismatch	CLASS_LABEL	Categorical	Phishing or legitimate (0,1)

Exploratory analysis







Important Variables

- Random Forest importance

Column name	Data type
PctExtHyperlinks	Numerical
PctExtNullSelfRedirectHyperlinksRT	Numerical
PctExtResourceUrls	Numerical
FrequentDomainNameMismatch	Categorical
PctNullSelfRedirectHyperlink	Numerical
NumDash	Numerical

Model and Parameter Selection

- Hyper parameter tuning
- Model selection
- Evaluating performance
- Optimizing cutoff value

Model Performance

Model	Recall	Accuracy
Logistic Regression	0.95	0.79
Support vector classifier	0.94	0.84
Neural network	0.75	0.80
Random Forest	0.98	0.92
Ada boost	0.98	0.95
Gradient boosting	0.98	0.96

Cutoff Values

Cutoff Value	Recall	Accuracy
.01	.99966	.61796
.05	.99279	.87831
.10	.98782	.92669
.15	.98294	.93471
.20	.9795	.94291
.25	.97576	.94612
.30	.97263	.94683
.35	.96954	.94714
.4	.96636	.94747
.45	.96156	.94695
.5	.95543	.94576

Best Models

- Voting Classifier
 - Recall: 99.2%
 - Accuracy: 94.9%

Model	n_estimators	max_depth	learning_rate
Random Forest	500	7	N/A
Ada boost	100	7	.01
Gradient boosting	2000	7	.1

Conclusion and Further Study

- Confidently identified phishing websites
 - 99.2% Recall
 - 94.9% Accuracy
- Different applications
 - Variables
 - Parameters

